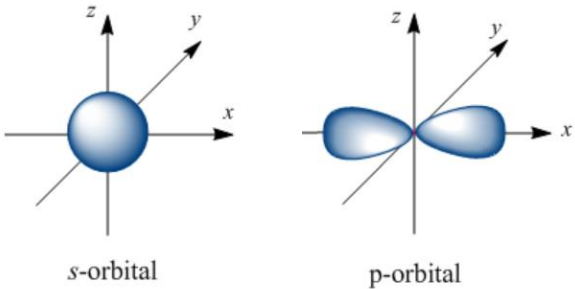
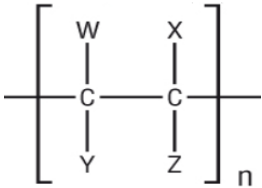


Section A

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
1.	(a)	(i)		orange to colourless	1			1		1
		(ii)		$\begin{array}{c} \text{H}_3\text{C}-\text{CH}-\text{CH}_2 \\ \quad \\ \text{Br} \quad \text{Br} \end{array}$ accept any correct structure for 1,2-dibromopropane accept bromo-hydroxy compound		1		1		
	(b)			(alkaline) potassium manganate(VII) (1) accept potassium permanganate	1			1		1
2.	(a)				1			1		
	(b)			π -bond	1			1		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
3.				add sodium carbonate / sodium hydrogencarbonate solution (1) fizzing/effervescence is observed (1) OR add magnesium / calcium (1) fizzing/effervescence is observed (1) OR add universal indicator (1) change colour (from green) to red (1)	2			2		2
4.						1		1		
5.				enthalpy change that happens when 1 mol of a substance/compound is formed (1) from it's constituent elements in their standard states under standard conditions (1)	2			2		
				Section A Total	8	2	0	10	0	4